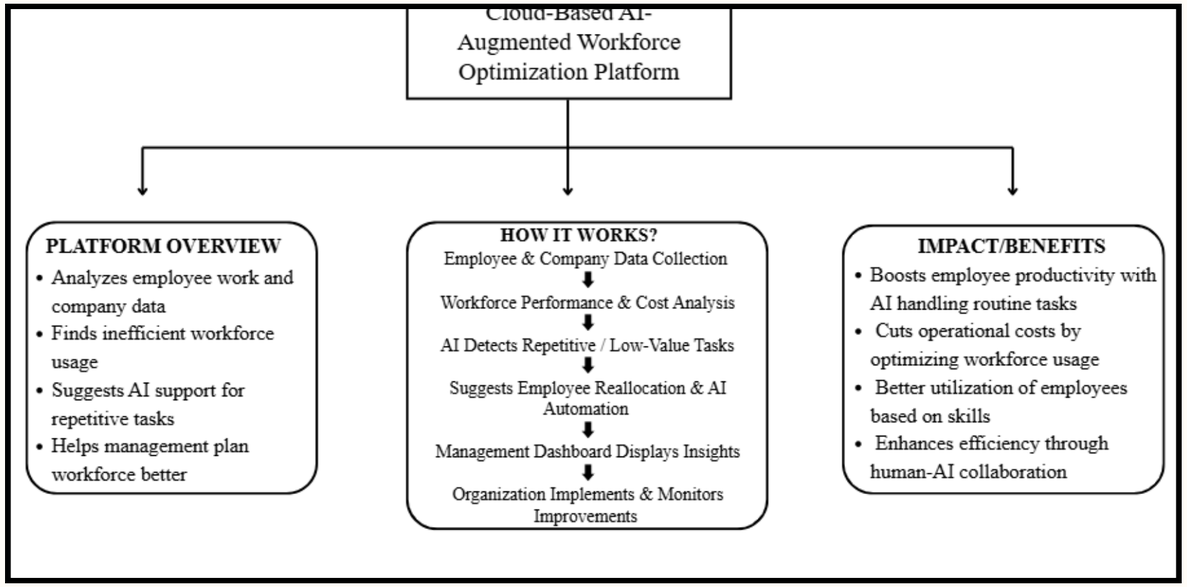
TOPIC: Workforce Cost Optimization & Productivity Analysis

PROBLEM STATEMENT

How can organisations improve profitability by efficiently reallocating employees and integrating AI tools to enhance productivity while reducing operational costs without reducing workforce strength?

PROPOSED SOLUTION



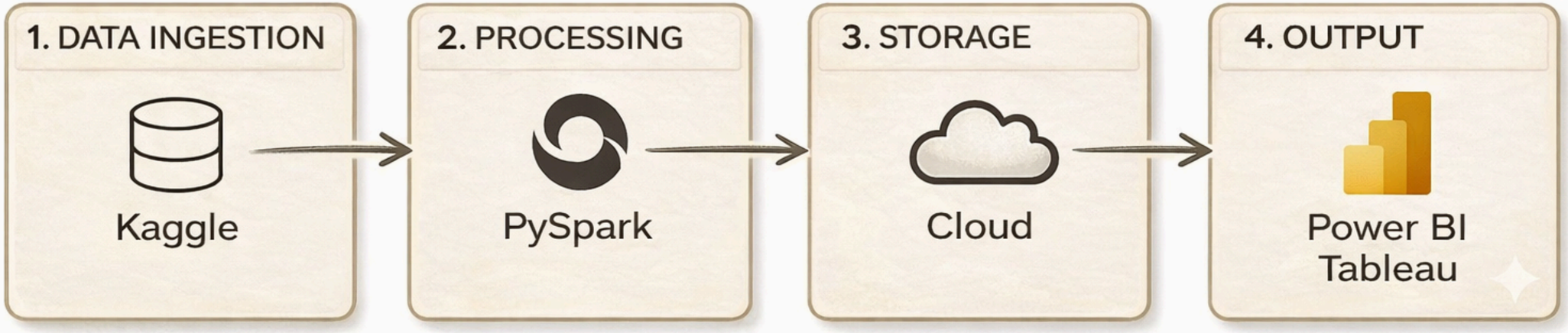
Objective

- To analyze employee data and identify factors affecting productivity.
- To understand workforce cost patterns across employees and departments.
- To detect inefficiencies in employee utilization using data-driven analysis.
- To support better workforce planning and cost optimization without reducing workforce strength.

GROUP MEMBERS:

1. Anjana M
2. Chanda
3. Kavita
4. Shivshankar

End-to-End Big Data Pipeline



1.Kaggle - Data Ingestion

We used Kaggle as the data source because it provides ready-to-use public datasets in a clean and accessible format. It is easy to download and use in Google Colab, which made the data collection step simple for our project.

2. PySpark – Processing

We used PySpark for data processing because PySpark is a big data tool used for handling and processing large datasets efficiently. It helps in performing operations like data loading, cleaning, transformation, and aggregation in a structured way.

3. Cloud – Storage

We included Cloud in the pipeline to represent where processed data can be stored for future use. In a real-world system, cloud storage is useful because it allows data to be stored centrally, accessed easily, and scaled when data volume increases.

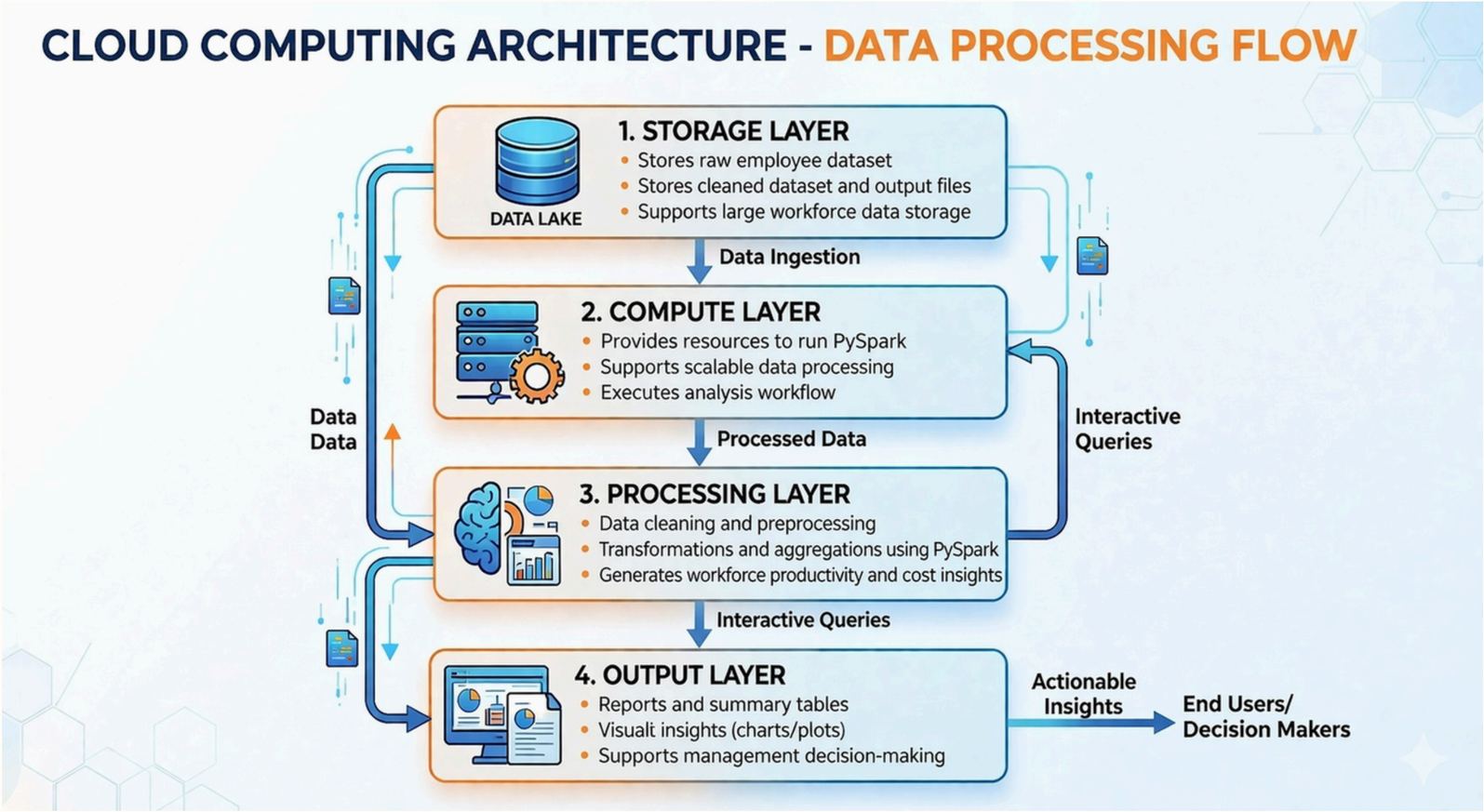
4. Power BI / Tableau – Output

We used Power BI / Tableau in the output stage because these tools help in converting processed data into easy-to-understand visualizations, charts, and dashboards. This makes the final insights clear and useful for decision-making.

Cloud Computing Architecture

Cloud Layers:

- Storage Layer → Stores large workforce data
- Compute Layer → Runs PySpark processing
- Processing Layer → Data cleaning and Preprocessing
- Output Layer → Dashboard & Reports



Cloud Computing Architecture - Explanation

Workflow

- The raw employee dataset is first stored in the storage layer.
- The compute layer provides the resources required to run the project and execute PySpark operations.
- The processing layer performs data cleaning, preprocessing, transformations, and aggregations on the dataset.
- The processed data is then passed to the output layer, where dashboards, reports, and visual insights are created.
- These final outputs help end users and decision makers understand productivity patterns and workforce cost issues.

Scalability

- The architecture is scalable because the storage layer can handle growing volumes of employee data.
- The compute layer can be expanded when more memory or processing power is required.
- The processing layer using PySpark supports efficient handling of larger datasets and more complex analysis.
- This makes the system suitable for real-world organizations where workforce data keeps increasing over time.